VALENTINE GOLDEN GHANEM

Principal Medical Laboratory Scientist | Epidemiology & Public Health Researcher

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ORCID: 0009-0002-8332-0220 | WOS: NRA-8276-2025

RESEARCH PROFILE

Senior Medical Scientist and Computational Epidemiologist with over 10 years of clinical laboratory leadership experience, specializing in the transition from diagnostic medicine to advanced predictive modeling. Professionally positioned at the intersection of medical microbiology, machine learning, and health law to translate facility-level diagnostic data (ISO 15189) into actionable models for global health equity. Demonstrated track record in engineering robust analytical pipelines utilizing ensemble machine learning algorithms (XGBoost, Random Forest) and spatial statistics (LISA) to forecast infectious disease trends and map sub-national healthcare disparities. Expert in leveraging national surveillance data repositories (DHIMS2) to strengthen health system resilience and guide data-driven public health interventions.

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PRIMARY RESEARCH INTERESTS

Epidemiology	Spatial Analysis of Infectious Diseases, Health Inequities, Social Determinants of Health.
Data Science	Machine Learning for Disease Forecasting (XGBoost, CART), Explainable AI (SHAP) in Public Health.
Global Health	Health Systems Strengthening, Global Health Law, Cost-Effectiveness of mHealth Interventions..

EDUCATION

Expected Dec 2026 — **Master of Laws (LL.M.) in International Law**
Liverpool John Moores University, UK
Thesis: *Pathogen Data Sovereignty and One Health Surveillance: Ghana's Legal Duties under Global Health Security Law*

Mar 2026 — **Master of Science (M.Sc.) in Public Health — Merit**
University of Suffolk, UK
Specialization: Epidemiology and NCD Control
Thesis: *Cost-effectiveness of mHealth care models for hypertension in Ghana.*

Dec 2025 — **Master of Science (M.Sc.) in Data Science — Distinction**
University of East London, UK
Specialization: Machine Learning and Spatial Analytics
Thesis: *Ensemble ML modeling for HIV/AIDS incidence forecasting.*

Nov 2013 — **Bachelor of Science (B.Sc.) in Medical Laboratory Science — Second Class Upper**
University of Ghana, Accra
Specialization: Medical Microbiology

SELECTED SCHOLARLY OUTPUT AND PUBLICATIONS

Peer-Reviewed Journal Articles

2026 Spatial and Machine Learning Analysis of District-Level Health Insurance Inequities in Ghana
Ghanem, V. G. (2026). Cureus, 18(1), e101984. 10.7759/cureus.101984
Impact: Bivariate LISA and CART identified district-level illiteracy as the primary predictor of uninsurance; isolated 42 high-risk clusters for targeted interventions.

2025 Spatial Determinants of HIV in Ghana: An Ecological Study
Ghanem, V. G. (2025). AIDS & Infectious Diseases, 15(4), 50–62. 10.4236/aid.2025.154050
Impact: Identified a 16% reduction in HIV incidence per 10% increase in education using SHAP/XGBoost; mapped regional hotspots via geospatial ML.

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2025 HIV Prognosis in Sub-Saharan Africa: A Narrative Review of Structural Mortality Drivers
Ghanem, V. G. (2025). International Journal of Medical and Health Research, 3(2), 45–53. 10.61424/ijmhr.v3i2.346
Impact: Synthesised 218 studies; showed only 43% of HIV interventions provide actionable policy recommendations; highlighted equity gaps in programming.

PROFESSIONAL APPOINTMENTS

2015 – Present **Senior Medical Scientist (Supervisory) & Epidemiology Researcher**
Cocoa Clinic (Ghana COCOBOD Medical Department), Accra, Ghana

- Orchestrated the integration of laboratory-confirmed disease surveillance metrics directly into the national District Health Information Management System (DHIMS2) framework.
- Engineered robust machine learning pipelines (XGBoost, Random Forest) utilizing sub-national data streams to quantify socio-behavioral and environmental drivers of infection.
- Supervised medical laboratory diagnostic operations for high-throughput units processing over 30,000 monthly samples under strict compliance with ISO 15189 Quality Management System.
- Managed mobile screening outreach initiatives providing preventive care, health impact metrics, and diagnostics to 4,000+ underserved individuals in regional clusters.

2014 – 2015 **Field Diagnostic Coordinator (Volunteer)**
Life Medical Mission, Accra, Ghana

- Coordinated community health outreach platforms, managing mobile laboratory infrastructure and executing rapid screenings for endemic conditions across 5,000+ beneficiaries.

SCIENTIFIC SKILLS & COMPETENCIES

Quantitative	Biostatistics, Machine Learning (Python/R), Spatial Analysis (QGIS, LISA), SQL.
Software	Tableau, Streamlit, DHIMS2, LIMS (Labware, Meditech), GSS Census Data Extraction, Plotly Dash, GitHub/GitLab.
Laboratory	Medical Microbiology, Infectious Disease Diagnostics, ISO 15189 QMS.

ACADEMIC HONOURS, MEMBERSHIPS & LICENSURE

Licensure & Registration

- CORU Registered Medical Scientist, Ireland (MS 063958)
- AHPC Licensed Medical Laboratory Scientist, Ghana (09007815)
- Registered Epidemiologist A (Netherlands Society for Epidemiology - VvE)

Fellowship

- Fellow, Royal Society for Public Health (FRSPH) (140437)

Professional Memberships

- Member: ACSLM, Ireland (12474) | GABMLS, Ghana (AS 1206) | Dutch Epidemiology Society (VvE)

Certification (In Progress)

- Certified in Public Health (CPH) — In progress (December 2026 Exam Preparation)

Institutional Honors:

- Reciepeint (2026), University of East London Alumni Achievement Awards (Diversity / Industry of the Future Categories)
- International Partnerships Representative, University of East London Alumni Advisory Board
- Recipien (20226)t, Unicaf Africa Prize for Academic Excellence, recognising outstanding academic performance and multidisciplinary research leadership during my MSc Public Health at the University of Suffolk.